surveillance missions. The Cold War (1947–1991) intensified these developments as both superpowers sought new methods of intelligence gathering without endangering pilots. Early jet-powered drones like the Ryan Firebee series (1950s) demonstrated extended range and higher altitude capabilities, becoming the foundation for subsequent reconnaissance UAVs.

During the Vietnam War (1955–1975), the United States pioneered the use of UAVs for real-time intelligence and photographic reconnaissance. The Ryan Model 147 “Lightning Bug” performed more than 3000 missions, capturing aerial images and assessing bomb damage deep within hostile territories. These missions proved the tactical and operational value of unmanned aircraft in modern warfare. Similarly, in Israel, the development of the IAI Scout (1970s–1980s) marked a breakthrough in UAV technology, introducing real-time video transmission, data telemetry, and electronic warfare capabilities. The Scout’s battlefield success established Israel as a global leader in drone innovation.

By the 1980s and 1990s, UAVs evolved into more compact, reliable, and sensor-rich platforms. The introduction of microprocessors and solid-state electronics allowed for the miniaturization of flight control systems, while inertial navigation systems (INS) and global positioning system (GPS) modules provided precision navigation and automated flight control. The RQ-2 Pioneer (1986), developed jointly by the US Navy and Marine Corps, exemplified these advancements, featuring live video transmission, GPS-based route tracking, and reliable remote operation. Similarly, the CL-289 (1990), a collaborative project between Germany and Canada, showcased jet-powered propulsion and terrain-following radar, further refining autonomous navigation capabilities.